28TH MAY, 2025

REGIONAL MARITIME UNIVERSITY

MARINE ELECTRICAL/ELECTRONIC ENGINEERING DEPARTMENT

BEE 27 / BME 27
MIDSEMESTER EXAMINATION

BEEE 207 – ROTATING ELECTRICAL MACHINES

INSTRUCTIONS: ***ATTEMPT ALL QUESTIONS*** **TIME ALLOWED: 45 MINS**

1(a) The star-connected rotor of a 3-phase induction motor has a resistance and standstill reactance of 0.4 Ω/phase and 2.5 Ω/phase respectively. The e.m.f. induced between the slip rings at standstill is 80 V, the stator being connected to normal supply voltage. Find the rotor current and power factor at standstill when the slip rings are *(i)* short-circuited *(ii)* joined to a star-connected resistance of 5 Ω/phase. **[6 Marks]**

(b) State four advantages of 3-phase induction motor. **[4 Marks]**

2(a) Which of these three-phase induction motors: squirrel cage induction motor and wound rotor induction motor has higher starting torque and why? **[3 Marks]**

(b) A 3-phase, 50 Hz induction motor has 8 poles. If the full-load slip is 2.5 %, determine (i) synchronous speed (ii) rotor speed and (iii) rotor frequency. **[3 Marks]**

(c) State the three d.c. motor characteristics and briefly describe any of them. **[4 Marks]**

3(a) List the four main principal components (parts) of d.c. machines under its construction and explain the function of any two of them. **[4 Marks]**

(b) A 250 V, 4-pole, wave-wound series motor has 782 conductors on its armature. It has armature and series field resistance of 0.75 Ω. The motor takes a current of 40 A. Find its speed and gross torque developed if it has a flux per pole of 25 mWb. **[6 Marks]**

$E_b = V - I_a(R_a + R_{se})$

VM

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